

Please check the examination details below before entering your candidate information

Candidate surname					Other names				
Centre Number					Candidate Number				

Pearson Edexcel Level 3 GCE

Mock Paper Set 4

Paper reference	8MA0/22
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Mathematics
Advanced Subsidiary
PAPER 22: Mechanics

You must have: Mathematical Formulae and Statistical Tables, calculator	Total Marks
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Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Unless otherwise indicated, wherever a value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 30. There are 4 questions.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.
- If you change your mind about an answer, cross it out and put your new answer and any working underneath.

Turn over ►

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1. A train is moving along a straight horizontal track between two points A and B .

In a model of the motion of the train

- at time $t = 0$ the train is moving with speed 20 m s^{-1} as it passes A
- the train then moves with constant speed 20 m s^{-1} for 12 s
- the train then decelerates at a **constant** rate $a \text{ m s}^{-2}$ for 8 s, reaching a speed of 16 m s^{-1}
- the train then accelerates at 0.4 m s^{-2} until it reaches B

Using the model,

- (a) sketch a speed-time graph for the motion of the train between A and B , (2)

- (b) find the value of a . (1)

The train travels 285 m while it is **accelerating** at 0.4 m s^{-2}

Using the model,

- (c) find the distance AB , (3)

- (d) find the total time taken for the train to move from A to B . (3)

- (e) State **one** limitation of the model of the motion of the train at time $t = 12 \text{ s}$. (1)



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Question 1 continued

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Question 1 continued

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Question 1 continued

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(Total for Question 1 is 10 marks)



2.

In this question you must show all stages of your working. Solutions relying entirely on calculator technology are not acceptable.

A particle P moves along the x -axis.

At time t seconds, $t \geq 0$, the velocity of P is $v \text{ m s}^{-1}$, where

$$v = 3t^2 - 8t + 5$$

- (a) Find the values of t for which P is instantaneously at rest. (2)
- (b) Find the total distance travelled by P in the interval $0 \leq t \leq 2$ (4)

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Question 2 continued

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(Total for Question 2 is 6 marks)



3. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal perpendicular unit vectors.]

A particle P of mass 2 kg is at rest on a smooth horizontal plane.

Two horizontal forces, $\mathbf{F}_1\text{ N}$ and $\mathbf{F}_2\text{ N}$, are applied to P , causing P to move with acceleration $(4\mathbf{i} - \mathbf{j})\text{ ms}^{-2}$

(a) Find the magnitude of the acceleration of P . (2)

(b) Find, in the form $(a\mathbf{i} + b\mathbf{j})\text{ N}$, the resultant of $\mathbf{F}_1\text{ N}$ and $\mathbf{F}_2\text{ N}$. (2)

Given that $\mathbf{F}_1 = (2c - 1)\mathbf{i} + (c + 1)\mathbf{j}$, where c is a constant,

(c) find $\mathbf{F}_2$ in terms of $\mathbf{i}$, $\mathbf{j}$ and c , fully simplifying your answer. (2)



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Question 3 continued

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(Total for Question 3 is 6 marks)



4.

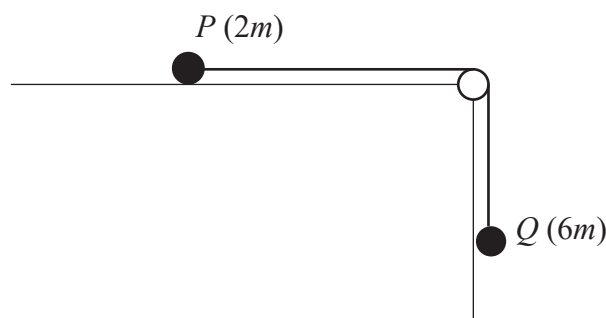


Figure 1

Two balls, P and Q , have masses $2m$ and $6m$ respectively.

The balls are attached to the ends of a rope that passes over a pulley. The pulley is fixed at the edge of a horizontal table.

Ball P is held at rest on the table, with the rope taut and Q hanging vertically below the pulley, as shown in Figure 1.

The system is released from rest and the balls move.

In a model of the motion

- the balls are modelled as particles
- the rope is modelled as being light and inextensible
- the pulley is modelled as being small and smooth
- air resistance is modelled as being negligible

In the motion

- Q moves downwards with an acceleration of magnitude $\frac{7g}{16}$
- the total resistance to the motion of P has magnitude kmg , where k is a constant
- the tension in the rope either side of the pulley is T

Using the model,

- write down an equation of motion for P , giving your answer in terms of T , k , m and g ,
(2)
- find the value of k .
(4)

In reality, the pulley would **not** be smooth.

- State how this would affect the tension in the rope.
(1)

The model is refined so that the pulley is modelled as **not** being smooth.

- Suggest **one** further refinement of the model, apart from including air resistance, that would make the model more realistic.
(1)



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Question 4 continued

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(Total for Question 4 is 8 marks)

TOTAL FOR MECHANICS IS 30 MARKS

